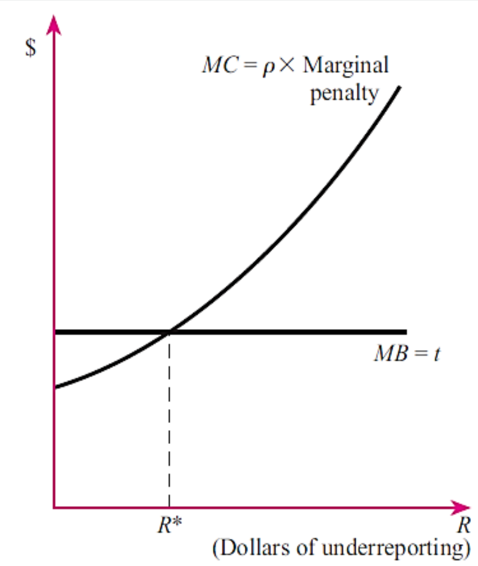
財政學 HW4

1. (Rosen 16.1) According to estimates by Goolsbee and Petrin [2004], the elasticity of demand for basic cable service is –0.51, and the elasticity of demand for direct broadcast satellites is –7.40. Suppose that a community wants to raise a given amount of revenue by taxing cable service and the use of direct broadcast satellites. If the community's goal is to raise the money as efficiently as possible, what should be the ratio of the cable tax to the satellite tax? Disscuss briefly the assumptions behind your calculation.

2. (Rosen 16.3) "Peter the Great at one time levied a tax upon beards. He held that the beard was a superfluous and useless ornament. The tax is said to have been proportional according to the length of the beard and progressive according to the social position of its possessor" [Groves, 1946, p. 51]. Evaluate Peter's beard tax from the standpoint of horizontal equity.

3. (Rosen 16.6) In 2008, New York State increased its tax on cigarettes by $1.25 per pack. One commentator noted that "most cigarettes sold [in New York] will actually be trucked up from Virginia or shipped from China, by 'butt-leggers' who can make over $1 million on each tractor-trailer load of smuggled smokes" [Fleenor, 2008]. Use Figure 16.5 to show how this behavior could have been predicted.

Figure 16.5
Tax evasion is positive
Given this tax rate, probability of being audited, and marginal penalty if caught cheating, the optimal amount of cheating is R^*, which is positive.



4. (Rosen 16.8) Indicate whether each of the following statements is true, false, or uncertain, and explain why:

(a) A proportional tax on all commodities including leisure is equivalent to a lump sum tax.

(b) Efficiency is maximized when all commodities are taxed at the same rate.

(c) Average cost pricing for a natural monopoly allows the enterprise to break even, but the outcome is inefficient.

(d) Tom's workplace provides free access to a fitness room; Jerry's does not. Horizontal equity requires that Tom be taxed on the value of having access to the fitness room.

5. (Rosen 16.5) Suppose that a town is considering a project to install new underground water pipes. Some of the costs are fixed in the sense that they do not increase with an increase in the amount of water consumed. For example, pipes deteriorate over time, independent of the volume of water flowing through them. Hence, it is impossible to pay for the investment in the pipes by charging consumers an amount based on the marginal cost of the water they consume. Under these circumstances, what might an efficient pricing system look like?

6. (Rosen 16.4) In 2008, the sales tax in Cook County, Illinois, was doubled. One journalist speculated that the tax increase was enacted because government officials sensed that the tax base was more captive than usual, since falling real estate prices and a tough job market made it difficult for citizens to move. Examine this situation through the lens of optimal tax theory. Is the increase in the sales tax efficient? Is it fair?